EXPERIMENT No.: 07

1. Aim of Experiment: To interface a Programmable Logic Controller (PLC) with a Human–Machine Interface (HMI) and monitor and control process variables in real time using an HMI.

2. Learning Outcomes

After performing this experiment, students will be able to:

1. Explain the basic architecture and working of an HMI/SCADA-based monitoring system.
2. Establish communication between a PLC and an HMI using a suitable communication protocol.
3. Configure tags to display real-time process data on an HMI screen.
4. Operate PLC outputs through HMI controls and verify the response of the system.
5. Identify industrial applications of HMI/SCADA in process monitoring and automation.

3. Components and Software Required

Sr. No.	Component / Software	Purpose
1	PLC	Executes the control logic and reads/writes process data.
2	HMI panel or HMI/SCADA software	Displays process information and accepts operator commands.
3	Communication cable or network	Connects the PLC and HMI.
4	PLC programming software	Creates and downloads the PLC program.
5	HMI configuration software	Designs the HMI screen and configures tags.
6	Power supply	Supplies power to the PLC and HMI as specified by the manufacturer.
7	Indicator lamp or equivalent safe output load	Demonstrates output control.

4. Basic Theory

4.1 Human–Machine Interface (HMI)

A Human–Machine Interface is a graphical interface that enables an operator to interact with an industrial machine or process. It presents information in a simple visual form using numerical displays, indicators, buttons, graphs, and alarm messages.

The operator can use an HMI to:

- Monitor process parameters such as temperature, pressure, speed, level, and motor status.
- Start or stop a machine through authorized control commands.
- Observe alarms and abnormal operating conditions.
- Change permitted operating setpoints.
- View trends and historical data, if supported.

4.2 Supervisory Control and Data Acquisition (SCADA)

SCADA stands for Supervisory Control and Data Acquisition. It is a supervisory system used to collect process data, present it to operators, generate alarms, and support remote monitoring and control.

A typical SCADA system may include PLCs, remote terminal units (RTUs), industrial communication networks, operator workstations, historical databases, and alarm management.

Difference between HMI and SCADA: An HMI commonly provides an operator interface for a particular machine or process. SCADA generally provides broader supervisory monitoring and control across multiple machines, processes, or geographically distributed installations. The functions may overlap.

4.3 Working Principle

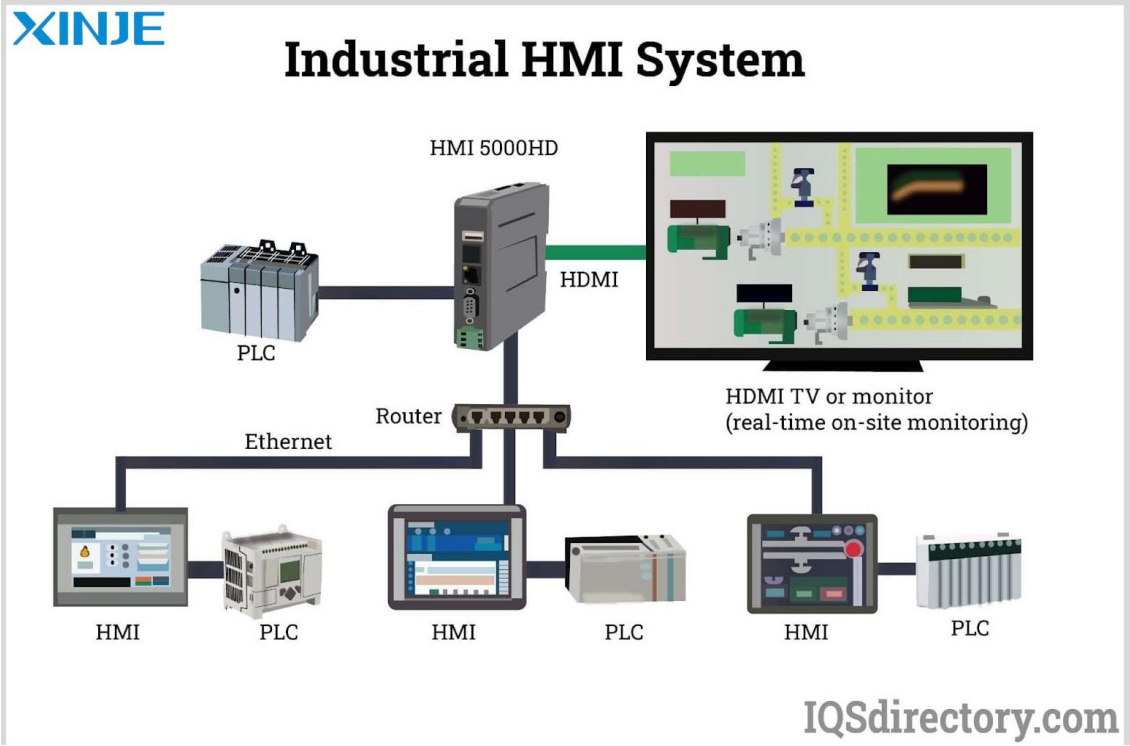
The PLC executes the control program and updates its input, memory, and output values. The HMI communicates with the PLC using a supported protocol such as Modbus TCP, Modbus RTU, PROFINET, or a manufacturer-specific protocol.

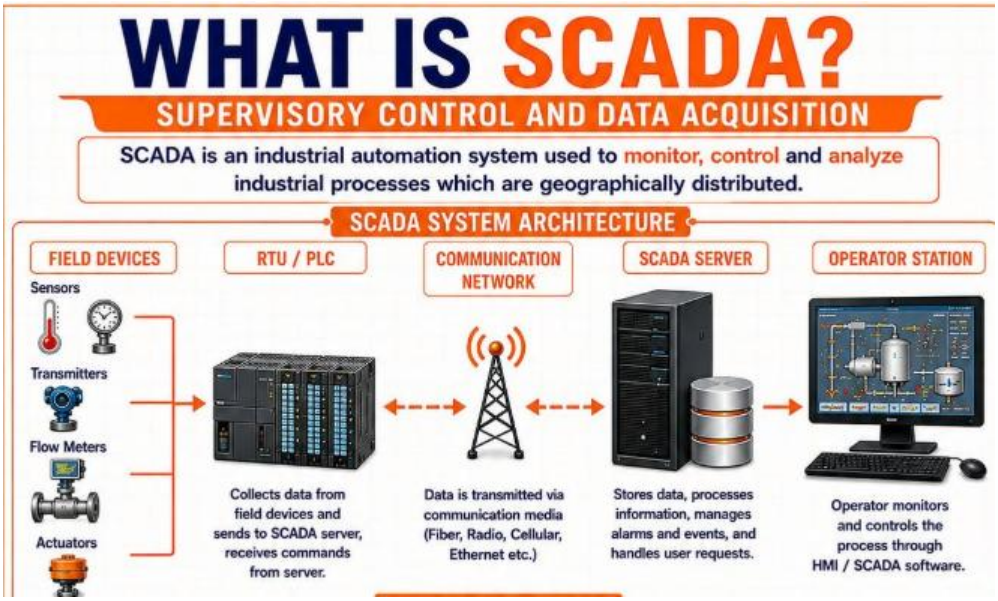
The HMI reads selected PLC variables and displays them on the screen. When an operator presses a control button, the HMI writes a command to a configured PLC variable. The PLC checks the command and executes the appropriate logic to operate the output.

Examples of monitored or controlled variables include:

- Motor ON/OFF status
- Tank level
- Temperature
- Conveyor operating status
- Pump command and feedback

5. Block Diagram





6. Experimental Setup

For demonstration, a simple motor and Pump control system is used.
The HMI screen contain the following objects:

1. Start button: Sends a start command to the PLC.
2. Stop button: Sends a stop command to the PLC.
3. Motor status indicator: Displays whether the motor or output is ON or OFF.
4. Pump status indicator: Displays whether the motor or output is ON or OFF.
5. Communication status indicator: Shows whether the HMI is communicating with the PLC, if supported.

The motor status must be based on PLC feedback or the actual output status, rather than merely copying the HMI command.



7. Sample PLC Tags

The following generic tags can be configured for the experiment. The addresses are illustrative; use the addresses and data types supported by the actual PLC.

8. Procedure

Part A: PLC Programming

1. Switch OFF the power supply before making electrical connections.
2. Connect the PLC to the programming computer using the appropriate cable or network.
3. Open the PLC programming software and create a new project.
4. Configure the PLC model and communication interface.
5. Create the required tags for start, stop, motor status, motor output, and process value.
6. Develop the PLC logic so that the start command energizes the output and the stop command de-energizes it.
7. Download the program to the PLC.
8. Verify the PLC program using simulation or a safe laboratory output.

Part B: Communication and Testing

1. Connect the PLC and HMI using the specified communication cable or network.
2. Check the device IP addresses or serial communication parameters.
3. Power ON the devices according to the laboratory instructions.
4. Download the HMI project to the panel or launch the configured HMI runtime.
5. Verify that the HMI communicates with the PLC without errors.
6. Observe the process value displayed on the HMI.
7. Press the Start button and observe the PLC output and corresponding indicator or safe test load.
8. Press the Stop button and verify that the output turns OFF.
9. Change the process value using the PLC program or a suitable test input and verify that the HMI display updates.
10. Record the observations and return the equipment to a safe condition.

10. Observation Table

Students should record the actual results obtained during the experiment.

Sr. No.	Test condition	HMI indication	PLC / output response	Result
1	PLC and HMI powered ON	Communication status displayed	PLC operational	____
2	Start button pressed	Motor status changes to ON	Output energized	____
3	Stop button pressed	Motor status changes to OFF	Output de-energized	____
4	Start button pressed	Pump status changes to ON	Output energized	____
5	Stop button pressed	Pump status changes to OFF	Output de-energized	____

11. Result

The PLC was interfaced with the HMI system using the supported communication interface. The configured process variables were displayed on the HMI, and the Start/Stop commands were tested to verify control of the PLC output. The experiment demonstrated the use of an HMI based system for process monitoring and operator control.

Write the result according to the actual observations obtained in the laboratory.

12. Industrial Applications

1. Process industries: Monitoring temperature, pressure, flow rate, and tank levels.
2. Manufacturing plants: Monitoring machine status, production count, and conveyor operation.
3. Water treatment plants: Monitoring pump operation, tank levels, and water flow.
4. Automobile industries: Supervising automated assembly lines and robotic work cells.
5. Power plants: Monitoring operating parameters and equipment status.
6. Food processing industries: Monitoring process temperature, filling operations, and production status.



13. Precautions

1. Verify all electrical and communication connections before switching ON the equipment.
2. Use the correct PLC model, communication driver, protocol, and device addresses.
3. Ensure that the PLC and HMI communication settings match.
4. Test output control with a suitable indicator or safe load before operating actual machinery.
5. Use proper interlocks and machine-safety provisions.
6. Do not change PLC program parameters while the equipment is operating unless the procedure permits it.
7. Verify that displayed values are current and valid; a frozen value should not be treated as real-time data.
8. Follow the manufacturer's instructions for wiring, power supply, and grounding.